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CNS Infections

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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1. Describe the specific pharmacological treatments for bacterial, viral, and fungal meningitis.
2. Describe the mechanisms of antimicrobials, mainly cell wall inhibitors, aminoglycosides, chloramphenicol, and rifampin. Include the mechanisms for resistance.
3. Describe the mechanism of antivirals which block viral DNA polymerase.
4. Describe the mechanism of antifungals, mainly azoles, flucytosine, and amphotericin.
5. Examine the specific adverse effects of these drugs. Include the management of these effects
6. Describe the role of glucocorticoids as supportive treatment. Include mechanisms and effects.

Source: Course Syllabus

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Microorganism	Standard Therapy	Alternative
Streptococcus pneumoniae	Third generation cephalosporin and vancomycin	Fluoroquinolone
Neisseria meningitides	Third generation cephalosporin	Chloramphenicol Fluoroquinolone Aztreonam
Listeria	Ampicillin or penicillin G; with aminoglycoside	SMX/TMP
H. influenzae	Third generation cephalosporin	Chloramphenicol Cefepime Meropenem Fluoroquinolone

Group B strep (babies) (*S. agalactiae*); *E coli* (neonates)—Use Beta lactams

*Vaccinate (start at age 11)

Meningitis: CSF values

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	Appearance	Opening Pressure mmHg	WBC (cell/μL)	Protein (mg/dl)	Glucose (mg/dL)
Normal	Clear	90-180	< 8	15-45	50-80
Bacterial Meningitis	Turbid	Elevated	>1000-2000	>200	<40
Viral Meningitis	Clear	Normal	<300; Lymphocytic predominance	<200	Normal
Fungal Meningitis	Clear	Normal-elevated	<500	>200	Normal - Low

Antimicrobials

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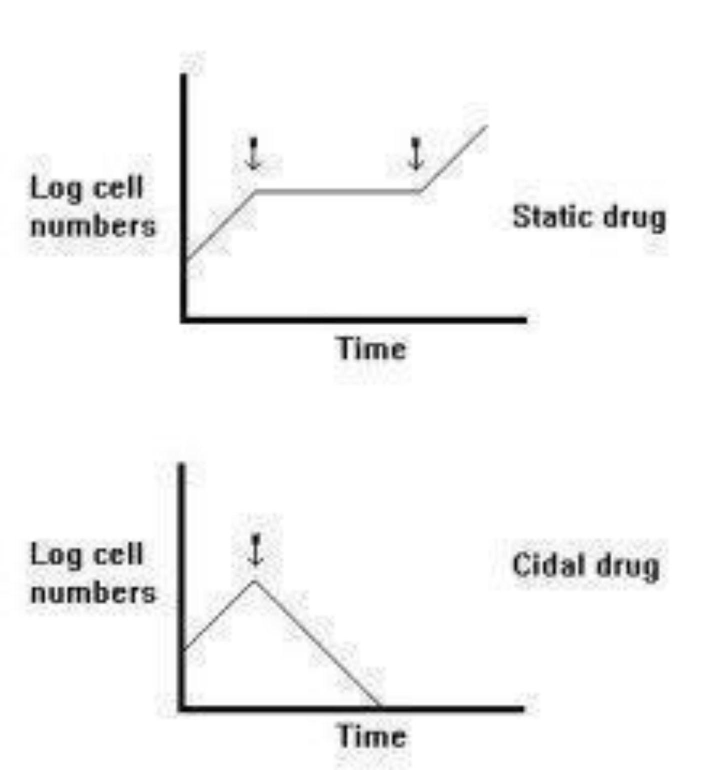
- Bacteriostatic
  - Inhibit the growth of bacteria
  - When used, the duration of therapy must be sufficient to allow cellular and humoral defense mechanisms to eradicate bacteria.
- Bactericidal**
  - Kills bacteria



Source: Course Syllabus

Classifications

- Broad Spectrum**
  - Covers both gram positive and gram negative
  - Initial management of meningitis (empiric therapy)*, plus bactericidal.
- Narrow Spectrum
  - Covers only gram positive

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TABLE 20.2	The Spectrum of Activity of Antibiotics and Other Antimicrobial Drugs							
Prokaryotes				Eukaryotes				
Mycobacteria*	Gram-Negative Bacteria	Gram-Positive Bacteria	Chlamydias, Rickettsias†	Fungi	Protozoa	Helminths	Viruses	
		←Penicillin→		←Ketocon-azole→		←Niclosamide→ (tapeworms)		
←Streptomycin	→				←Mefloquine→ (malaria)			
							←Acyclovir→	
						←Praziquantel→ (flukes)		
	←	Tetracycline	→					
←Isoniazid→								

*Growth of these bacteria frequently occurs within macrophages or tissue structures.
†Obligately intracellular bacteria.

The Cell Wall

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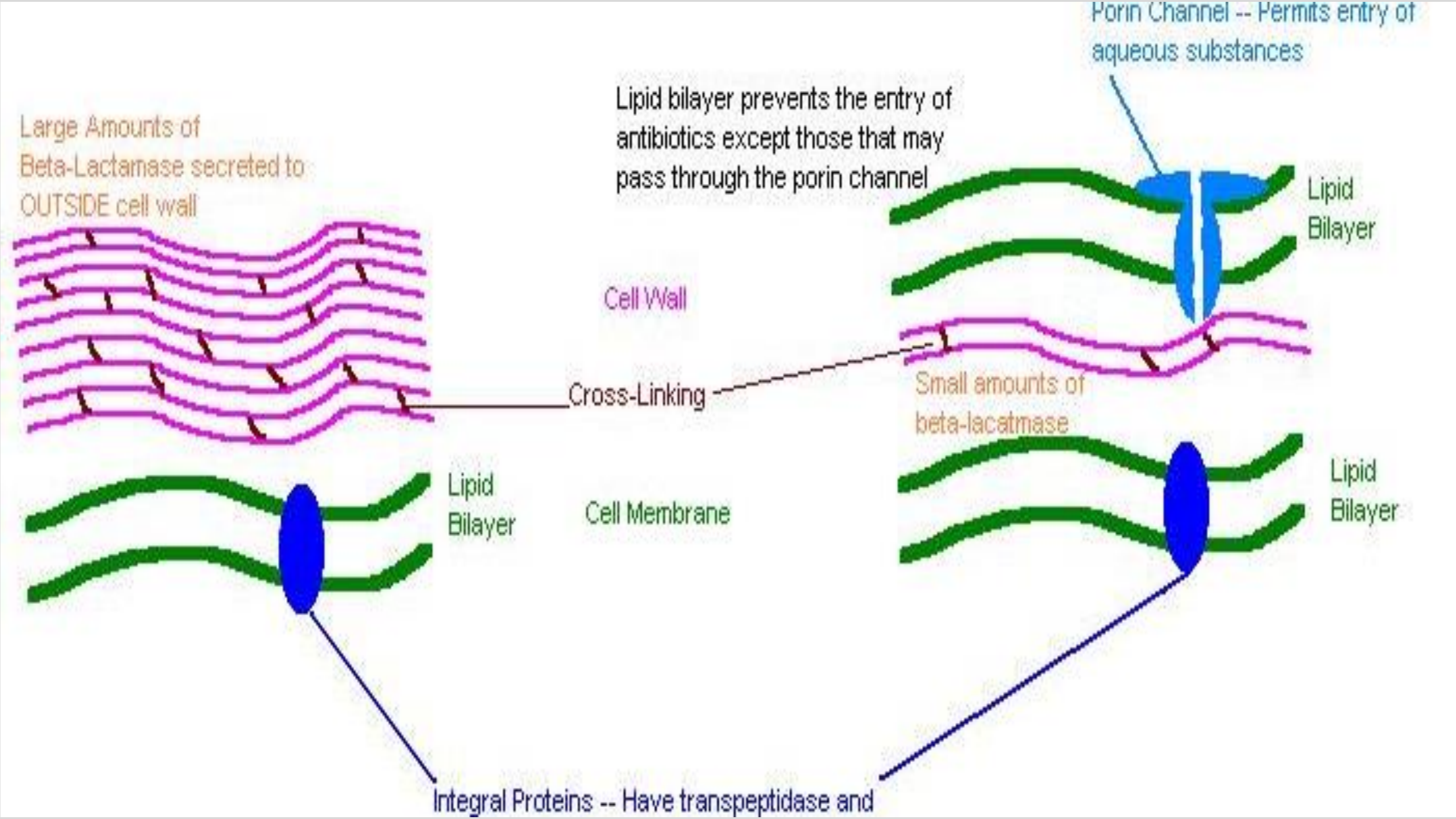

Cell Wall Inhibitors

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- **Bactericidal. Stops Transpeptidation**

  B- Lactams

  Penicillins: most are Gram +, some G-

  Cephalosporins: Early drugs are G+, later G+/-

  Monobactams (Aztreonam): Gram – (Porin allows bypass of cell surface, therefore no β-lactamase resistance)

  Carbapenems (Imipenem and cilastatin): Broad Spectrum

  Non B- Lactams

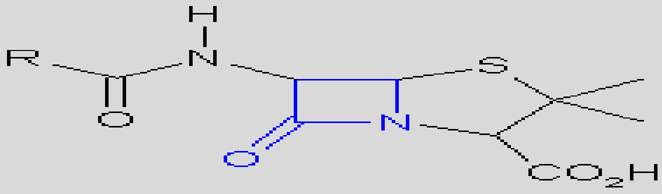
  Vancomycin: Gram +

  Daptomycin: Gram +

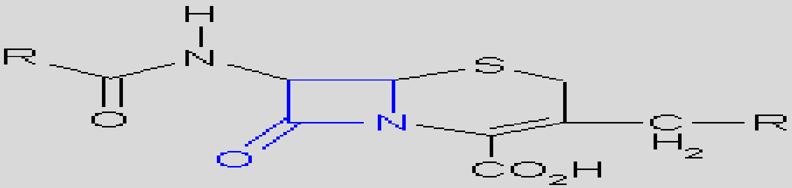
Source: Course Syllabus

The Beta Lactam Ring

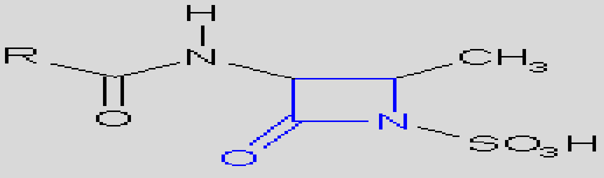
- These drugs share the β-lactam ring.
- Penicillinase (β-lactamase) disrupts the ring and is a major method for drug-resistance by bacteria.



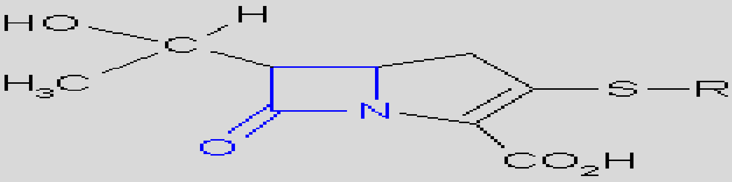
Penicillin nucleus



Cephalosporin nucleus



Monobactam nucleus

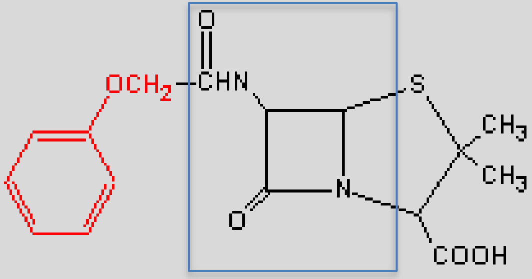


Carbapenem nucleus

Source: Course Syllabus

Resistance

- Degradation by beta lactamases (use “suicide” inhibitors)



- Mutation of penicillin binding proteins (methicillin)
- Porin down-regulation (gram - negative)
- Up-regulation of efflux channels

Source: Course Syllabus

Penicillin Uses

- Penicillin G (gram +, *Strep, Treponema*)
- Nafcillin, dicloxacillin, oxacillin (*S. auerus*)
- Ampicillin (both +/-, *Listeria)*
- Amoxicillin (both +/-, otitis media, sinusitis, *H pylori*, GU infections, *H influenzae, Strep*)
- Pipercillin, ticarcillin, carbenicillin (*Pseudomonas*)

Source: Course Syllabus

Penicillin Adverse Effects

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- Anaphylaxis
- Allergy to penicillins (1-2 % cross reactive with cephalosporins)
- Pseudomembranous colitis (ampicillin)
- Probenecid decreases excretion
- Interstitial nephritis
- Drugs which share penicillin hypersensitivity: Penicillins, Cephalosporins, Carbapenems. Similarity in the side chain.
- Drugs which don't: Aztreonam, Vancomycin

Source: Course Syllabus

Carbapenems

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- Meropenem, doripenem, etrapenem
- No cilastatin needed (unlike imipenem)
- Etrapenem longest $t_{1/2}$, but irritating (administer with lidocaine)
- > action against gram-negative aerobes vs gram- positive
- Renal elimination (renal failure raises seizure risk)

Cephalosporins

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- MOA: as penicillin.
- Inadequate distribution to CNS; except later generations
- Adverse effects: as penicillin; small possibility of cross-reactivity with hypersensitivity
- Drugs with methylthioterazole ring may produce bleeding abnormalities and disulfiram-like reactions.

Source: Course Syllabus

Cephalosporins

MOA and adverse effects: As penicillin

1-2% cross reactivity with penicillin

Generation	Drug	Use
First	Cefazolin (IV) Cephalexin (po)	mostly gram + infections
Second	Cefuroxime (IV, po) Cefoxitin (IV)	*E. coli, Klebsiella, Proteus, H influenzae,* more gram – than the first
Third	IV; Ceftriaxone, Ceftazidime	Most gram – and gram +, BBB penetration
Fourth	IV, Cefepime	As third generation,more resistant to beta lactamase

Vancomycin

- Inhibits cell wall synthesis by binding 2 d-alanine.
- Bactericidal; gram-positive
- Oral (*C diff*), IV (MRSA, Enterococci)
- Adverse effects: Ototoxicity, Red Man's syndrome (infuse slowly plus use diphenhydramine), Nephrotox.
- Vancomycin-Resistant Enterococci (VRE) happened. D-ala was replaced with D-lactate. What will you use, then???

Source: Course Syllabus

Cell Wall Synthesis

L-alanine → D-alanine

Alanine Racemase

Cycloserine

vancomycin

Glycopeptide Polymer + 2 D-alanine

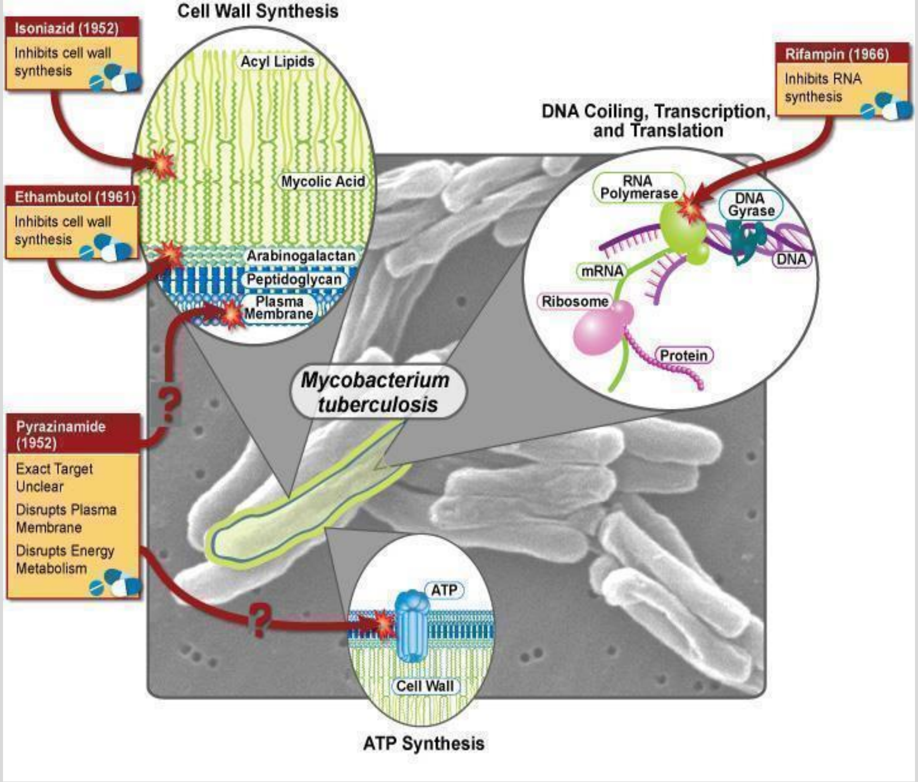
Transpeptidase (penicillin, cephalosporin, carbapenem)

(Penicillin-Binding Proteins)

Cross-linked Proteoglycan Complex

Rifampin

- **Inhibits DNA dependent RNA polymerase**
- ***Mycobacterium tuberculosis,* meningococcal prophylaxis.**
- **Adverse effects: Inducer of CYP, hepatotox, and red-orange secretions.**
- **Resistance: altered structure of enzyme.**



Source: Course Syllabus

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Aminoglycosides

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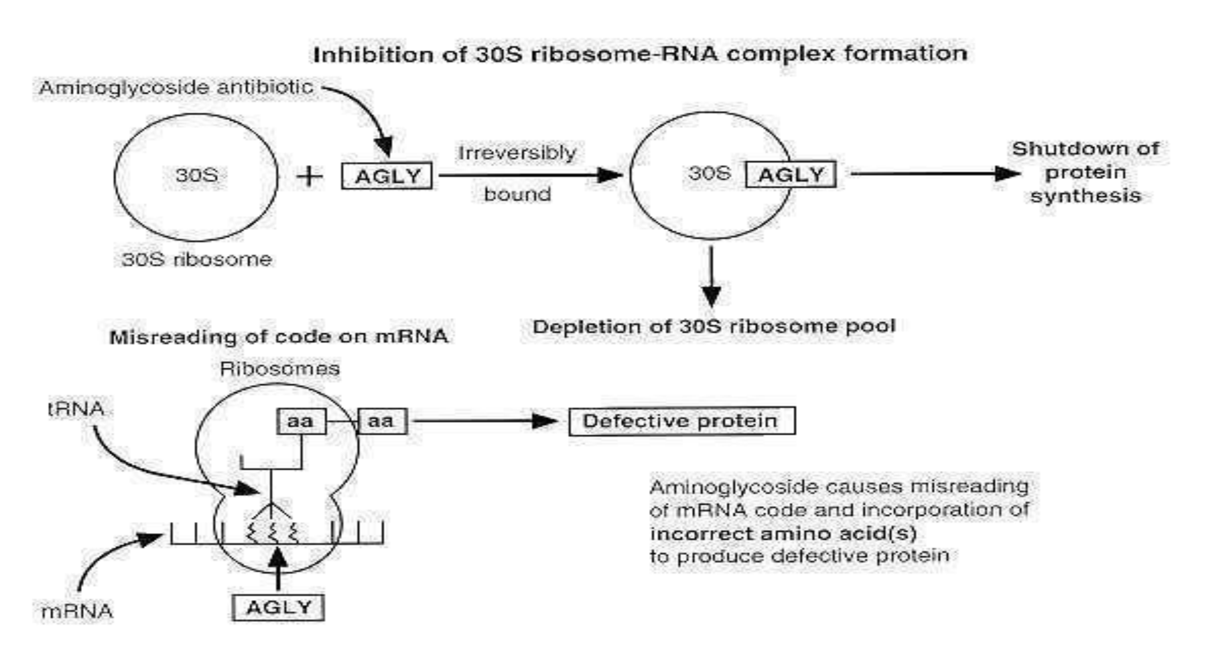
- Gentamicin
- Neomycin
- Amikacin
- Tobramycin
- Streptomycin

"GNATS"



Source: Course Syllabus

Aminoglycoside Resistance

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- Ribosomal mutation
- Decreased uptake of drug
- Inactivation of drug by enzymes
  - Acetyltransferases
  - Adenyltransferases
  - Phosphotransferases

Source: Course Syllabus

Aminoglycoside Adverse Effects

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- <u>Neuromuscular Blockade</u>
- Decreases acetylcholine release from the pre-synaptic neuron
- Avoid in anesthesia with neuromuscular blockade, myasthenia, and Parkinsonism.

- <u>Ototoxicity</u>
- Toxic to both hearing and balance.
- Overlap toxicity with another ototoxic drug (loop diuretics, cisplatin)
- Cranial nerve 8 damage (pregnancy)

Source: Course Syllabus

Aminoglycoside Adverse Effects

- <u>**Nephrotoxicity**</u>
- **Drug concentrates in proximal tubule.**
- **May cause Acute Tubular Necrosis (ATN). Loss of urine concentrating ability. Non-oliguric renal failure.**
- **Caution in individuals taking other nephrotoxic drugs***
- **Post antibiotic effect: enables once a day dosing.**

Source: Course Syllabus

Chloramphenicol

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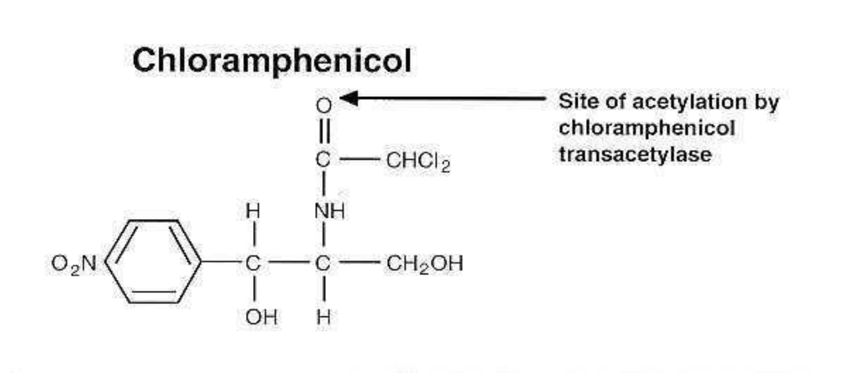
- Broad Spectrum
- Bacteriostatic
- Inhibits peptide formation by binding peptidyltransferase on 50S ribosome



Source: Course Syllabus

- Brain abscess
- Meningitis (*H influenzae, S pneumoniae, N meningitidis*)
- Typhoid
- Rocky Mountain Spotted Fever (*Rickettsial* meningitis doxycycline first line)
- Elimination by glucuronidation
- Wide Distribution: CSF, tears, placenta
- Reduce dose in liver failure.

Chloramphenicol

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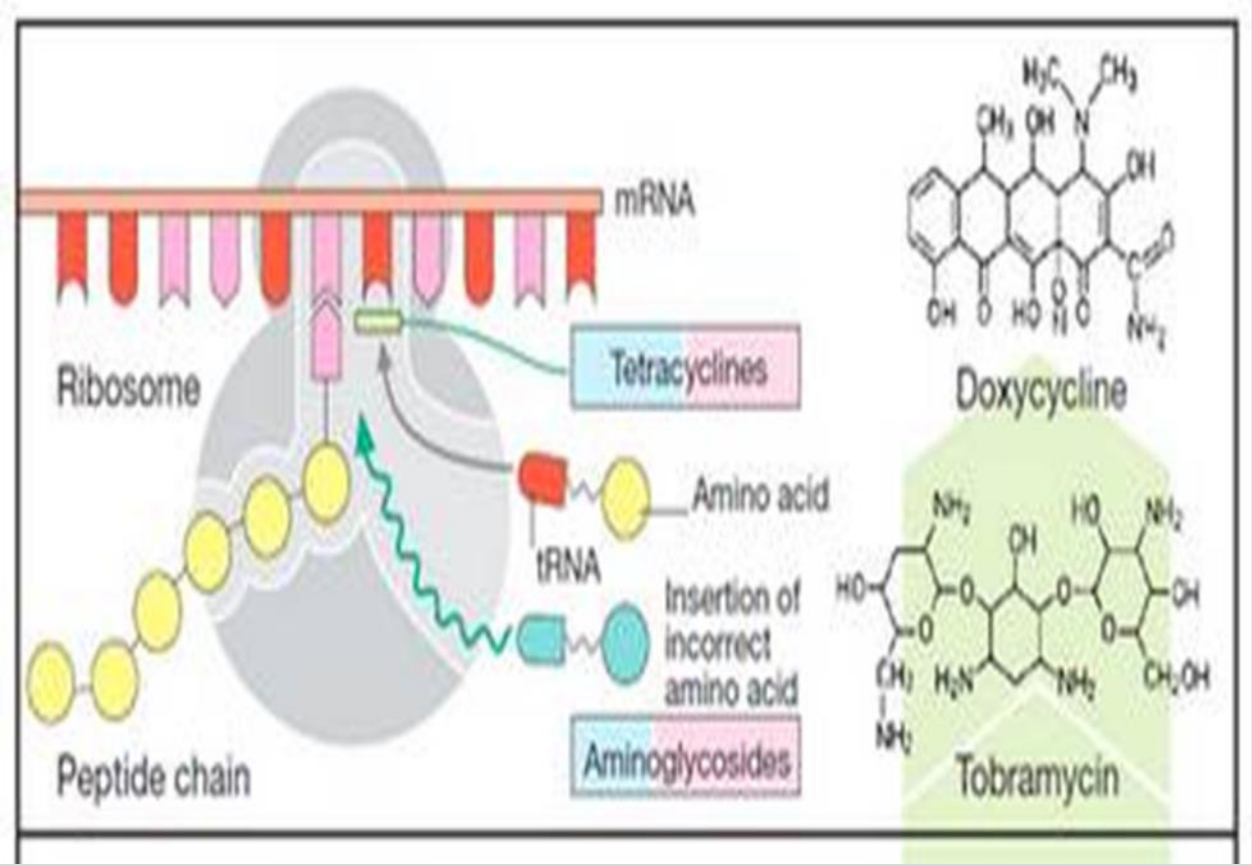
- <u>**Adverse Effects:**</u>
- **Bone marrow suppression**
- **Aplastic anemia**
- **Gray Baby Syndrome**

- <u>**Resistance:**</u>
- **Bacteria are impermeable to drug**
- **Acetylation of drug**

Source: Course Syllabus

Tetracyclines



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Tetracyclines inhibit "ELONGATION"

Tetracyclines

- Tetracycline
- Doxycycline (*Borrelia, Rickettsial* meningitis)
- Minocycline
- Tigecycline: (treats MRSA not sensitive to vancomycin)
- Demeclocycline (treats SIADH)

Elimination: Renal (doxycycline- fecal)

Distribution: Crosses BBB, placenta, fetal bones, breast milk, and teeth may accumulate drug.

Tetracycline Activity

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- Acne
- Rickettsial and Chlamydial infections
- *Mycoplasma*
- Cholera
- Brucella
- Lyme disease (*Borrelia burgdorferi*)
- Mycobacterium marinum
- *H pylori*

Tetracycline Adverse Effects: Teeth

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Teratogen. Not for pediatric or pregnant patients

Tetracycline Adverse Effects

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- **GI irritation**
- **Photosensitivity**
- **Increased azotemia**
- **Loss of renal proximal tubule function; Fanconi Syndrome**
- **Vertigo. Mainly minocycline.**
- ***C. difficile* infections**
- **<u>Resistance</u>: decreased drug uptake into cells or efflux out**

Fluoroquinolones "floxacins"

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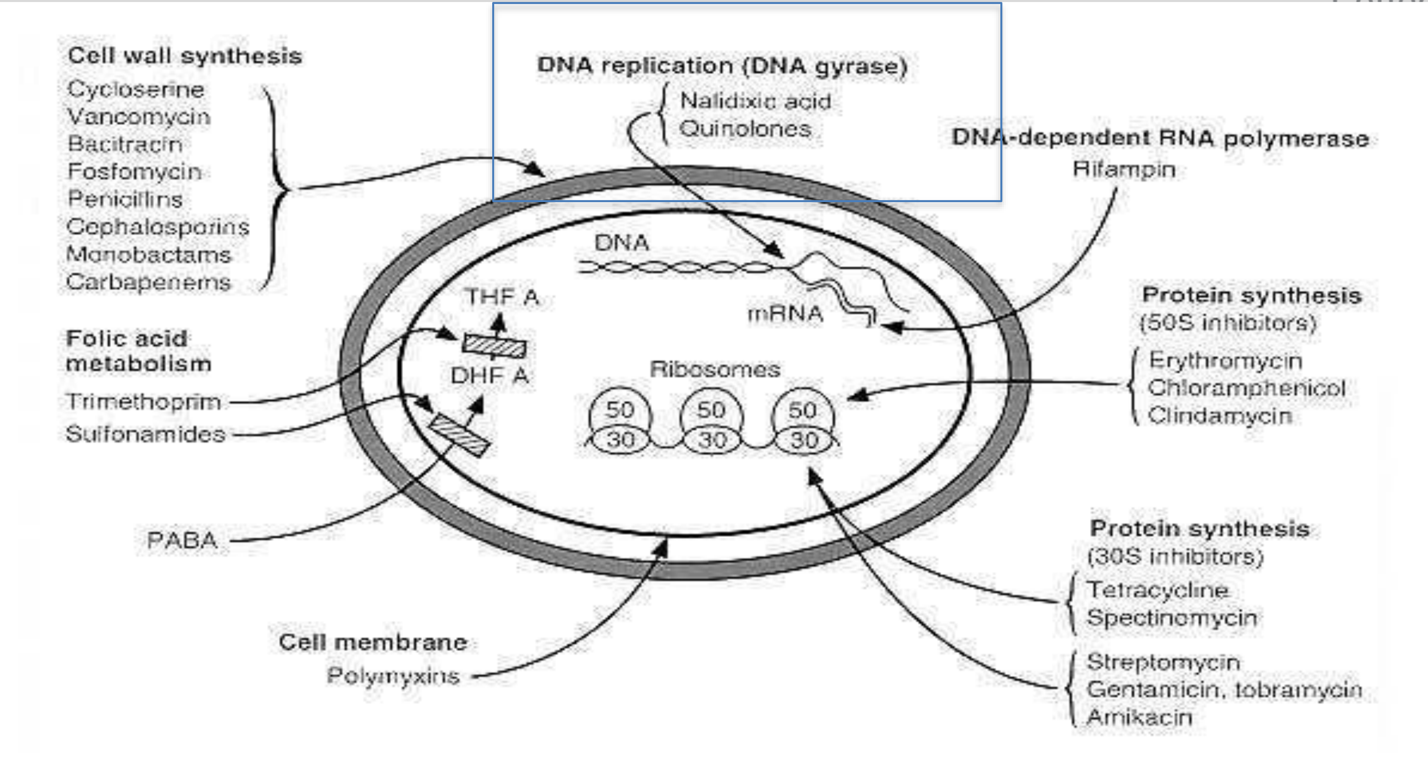

Source: Course Syllabus

Fluoroquinolones “floxacins”

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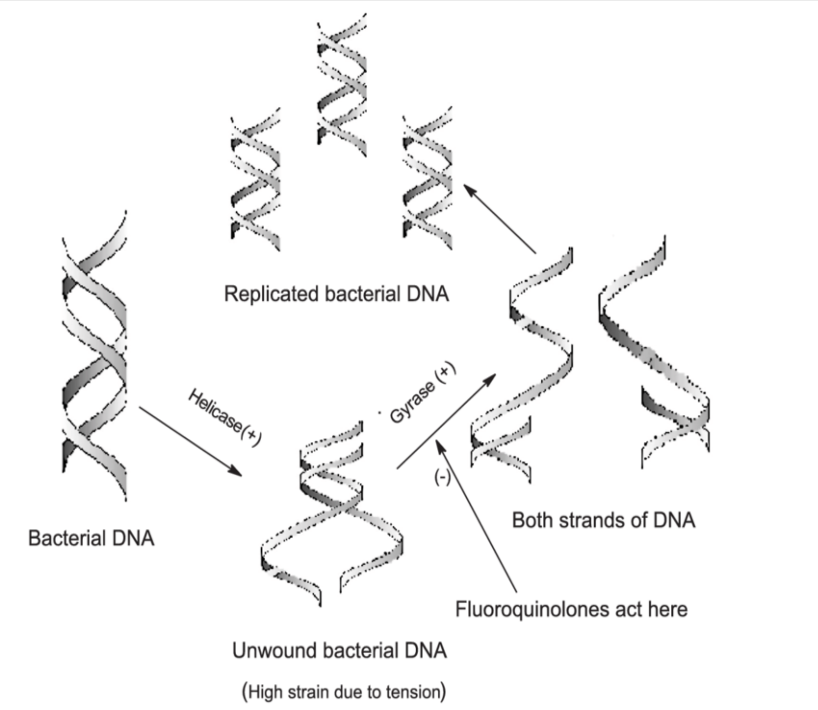

- Use: Gram – rods of urinary and GI tract, *Pseudomonas, Neisseria, Shigella, E coli, Proteus, Klebsiella, Anthrax* and some gram + infections. Bactericidal
- Block DNA gyrase and topoisomerase IV.
- Inhibits the negative supercoiled configuration of the bacterial chromosome during and following DNA replication.

https://www.researchgate.net/figure/Schematic-representation-of-mechanism-of-action-of-fluoroquinolones_fig1_40849381

Fluoroquinolones

- AE: Increased QT interval
- Damage growing cartilage
- Category C
- Tendon rupture
- Photosensitivity
- Drug interactions: CYP 1A2, di, trivalent cations
- Resistance: Reduced uptake of antibiotic, alteration of DNA gyrase

- Ciprofloxacin: broad spectrum
- Levofloxacin, Moxifloxacin
- “Respiratory” *Legionella, Mycoplasma, Chlamydia, Klebsiella, Mycobacterium avium-intracellulare, H. influenzae, S. pneumoniae.*

Source: Course Syllabus

Viral Meningitis

- Herpes Simplex I
- St. Louis
- West Nile
- Equine
- Measles, Mumps, Influenza
- Polio

<u>**Treatments:**</u>

- Acyclovir/Valcyclovir: HSV
- Ganciclovir: CMV
- Neuraminidase Inhibitors: Influenza
- Steroids
- Supportive

Source: Course Syllabus

Viral Meningitis: Enterovirus

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- 85% of all viral meningitis
- Polioviruses, Coxsackieviruses A and B, and Echoviruses
- Enter via oral-fecal route, or respiratory
- Management is supportive
- Concern for elderly and immunosuppressed

Source: Course Syllabus

Viral Meningitis: Arbovirus

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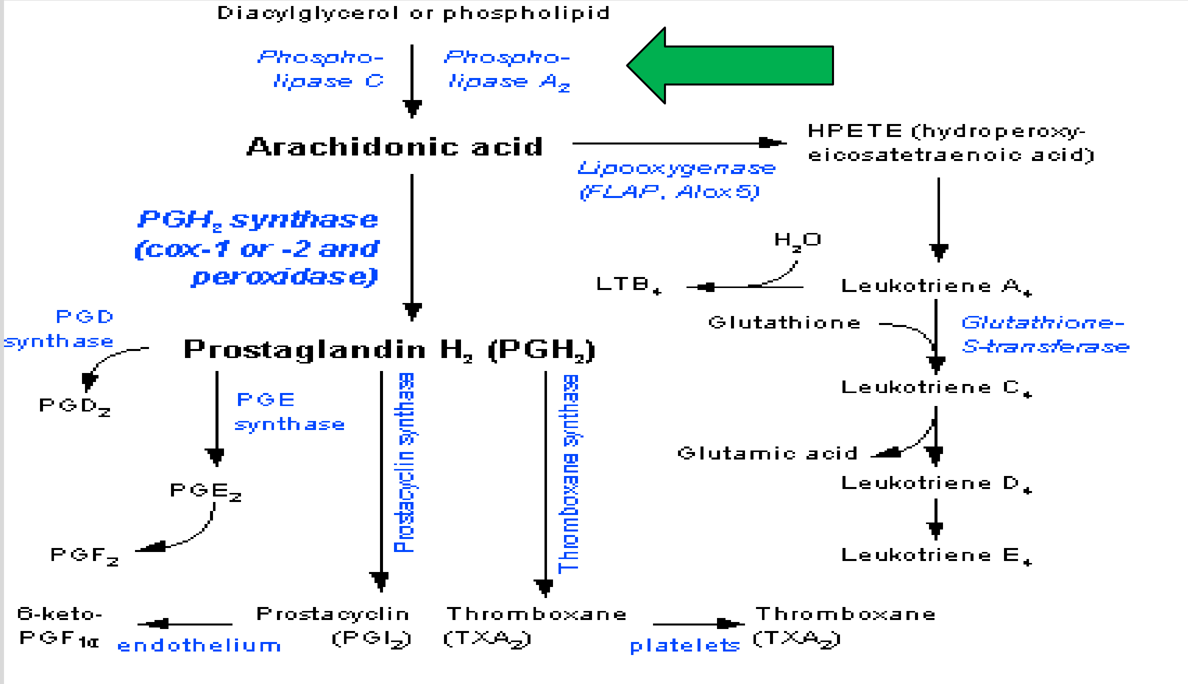
- 5% of meningitis cases.
- Eastern/Western/Venezuelan equine, St. Louis, California, West Nile, Zika, Dengue.
- Blood-sucking arthropods are the vectors.
- Insect repellant use recommended
- Seizures, Meningocephalitis.
- Treatment is supportive, suppress brain swelling with corticosteroids.

Source: Course Syllabus

Glucocorticoids

- Stimulate synthesis of lipocortin
  - Lipocortin inhibits Phospholipase A_2
    - Blocks starting point for prostaglandin, leukotriene synthesis



Glucocorticoids

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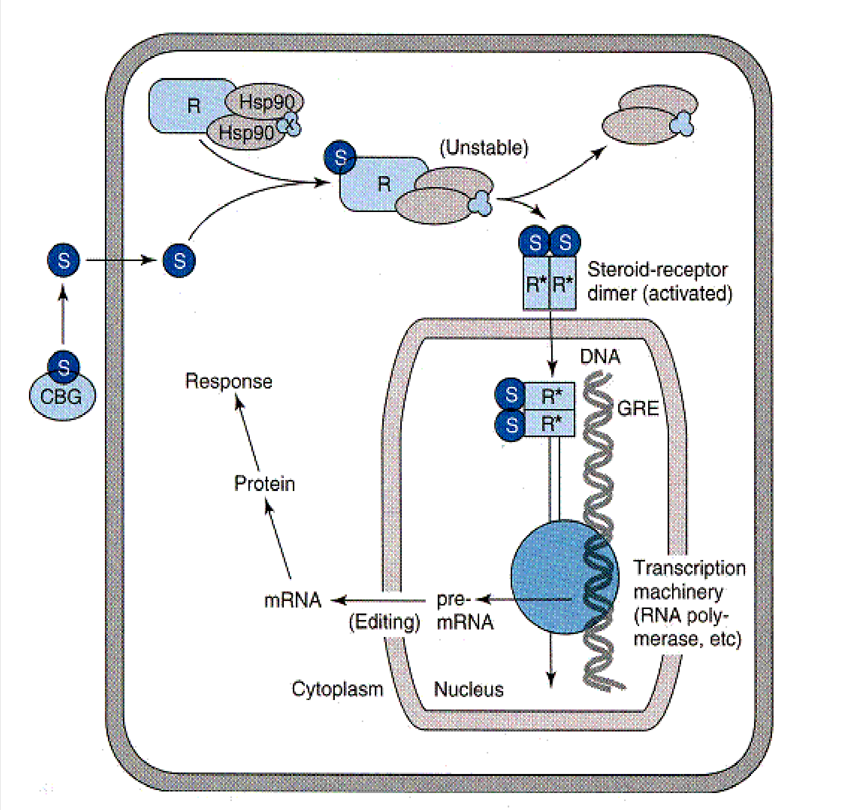
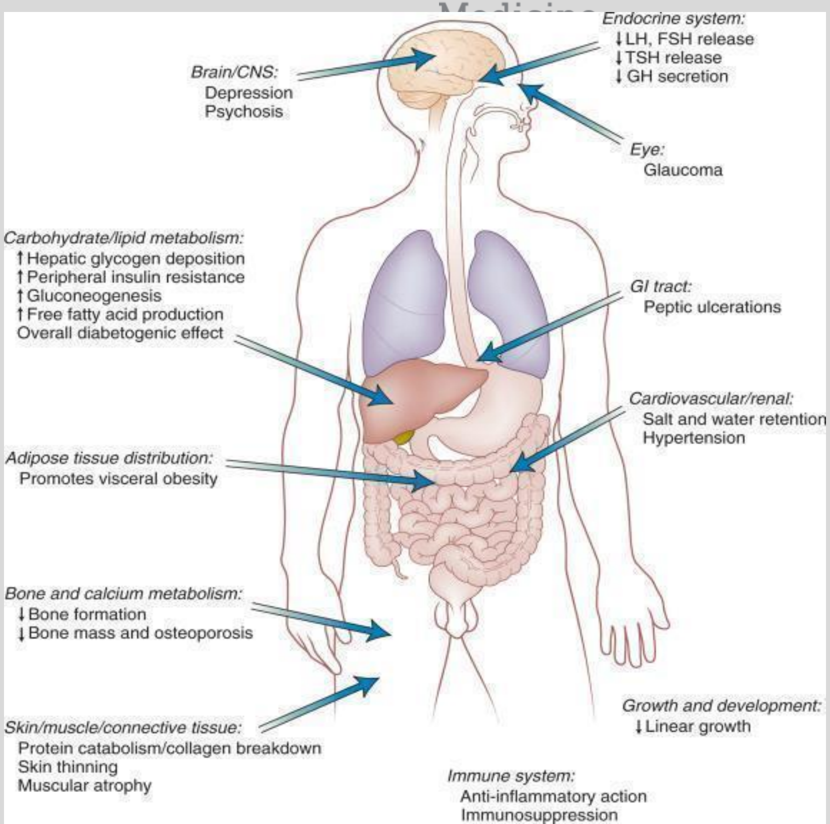
HPA suppression: taper dose slowly





Viral Meningitis

- **Mumps causes 20–40% of meningitis cases in undeveloped countries.**
- **Influenza, mumps, measles: Vaccinate**
- **Influenza: Zanamivir, Oseltamivir. Neuraminidase Inhibitors**

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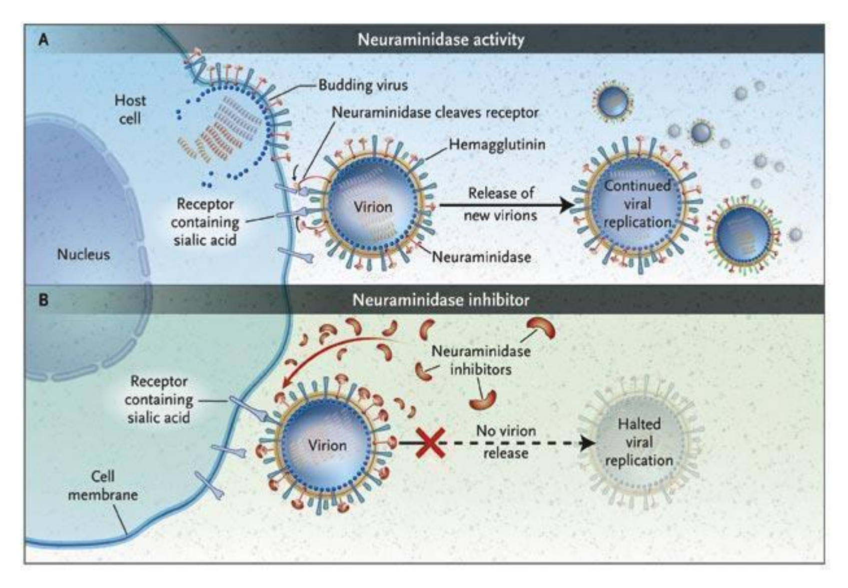

Source: Course Syllabus

Herpes

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- 4% of viral meningitis
- HSV-2 is most common cause.
- Herpes simplex, Varicella-zoster, Epstein Barr, Cytomegalovirus (CMV)
- Acyclovir is beneficial for most cases
- HAART therapy for HIV patients *
- Ganciclovir reserved for severe CMV infections

Source: Course Syllabus

Viral DNA Polymerase Inhibitors

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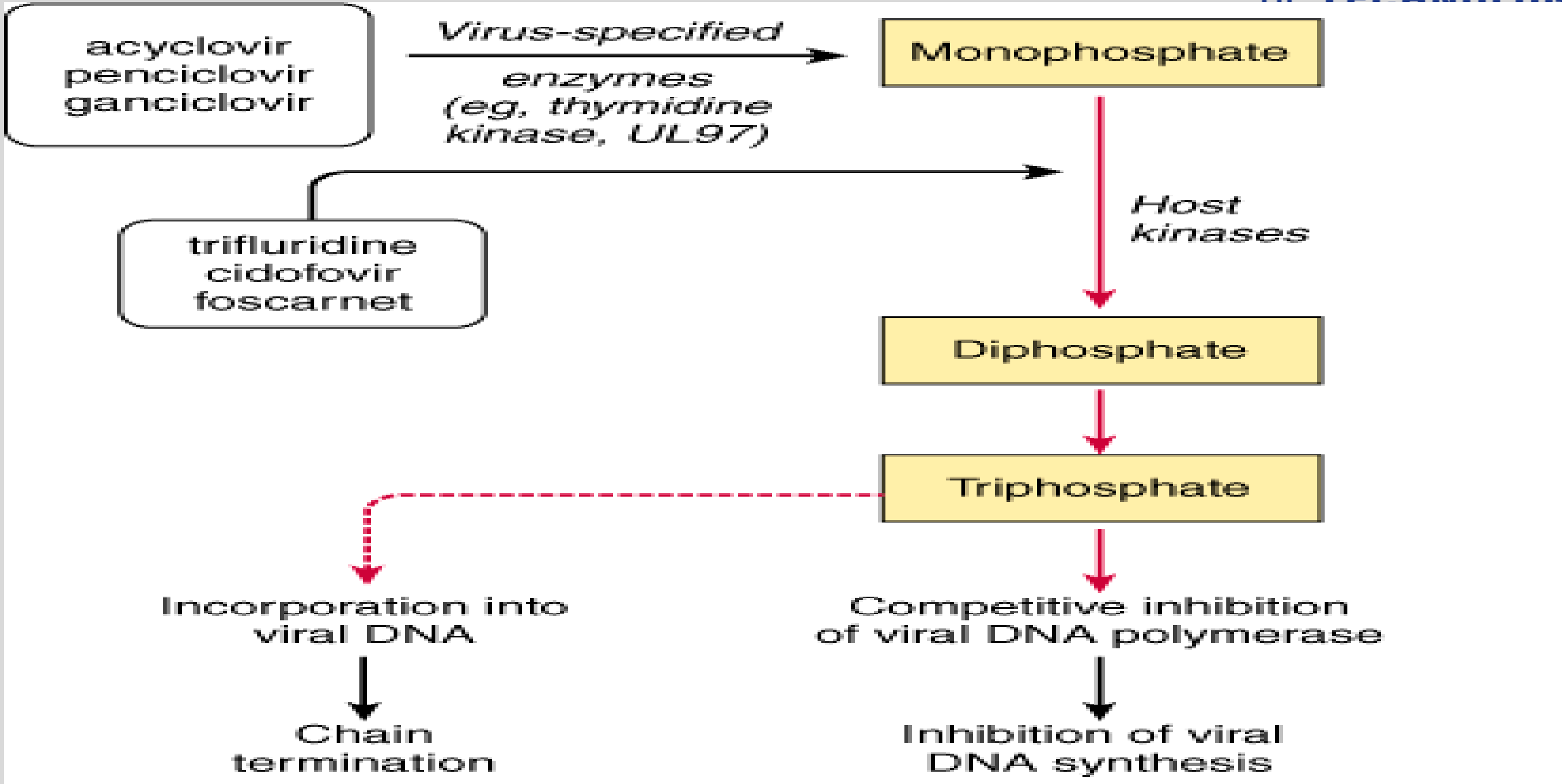

Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

Acyclovir

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- IV or topically (Oral HSV, VZV)
- IV use: HSV encephalitis, 1st and recurrent genital HSV, serious HSV or VZV, transplant surgery: HSV reactivation (50%)
- Adverse effects: Nephrotoxicity includes crystallized acyclovir (give hydration), seizures. Valcyclovir (prodrug)
- Foscarnet: phosphorylated; activation not needed (better for resistance). Hypocalcemia, nephrotoxicity.

Source: Course Syllabus

Ganciclovir

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- Guanosine analog
- Poor bioavailability. Given IV or use valganciclovir
- Uses:
- Cytomegalovirus (CMV) retinitis
- Kaposi's Sarcoma prevention in AIDs.
- 100x more effective than ACV vs CMV
- Adverse effects: Bone marrow suppression
- Resistance: Mutated CMV DNA polymerase or lack of viral kinase. Use foscarnet.

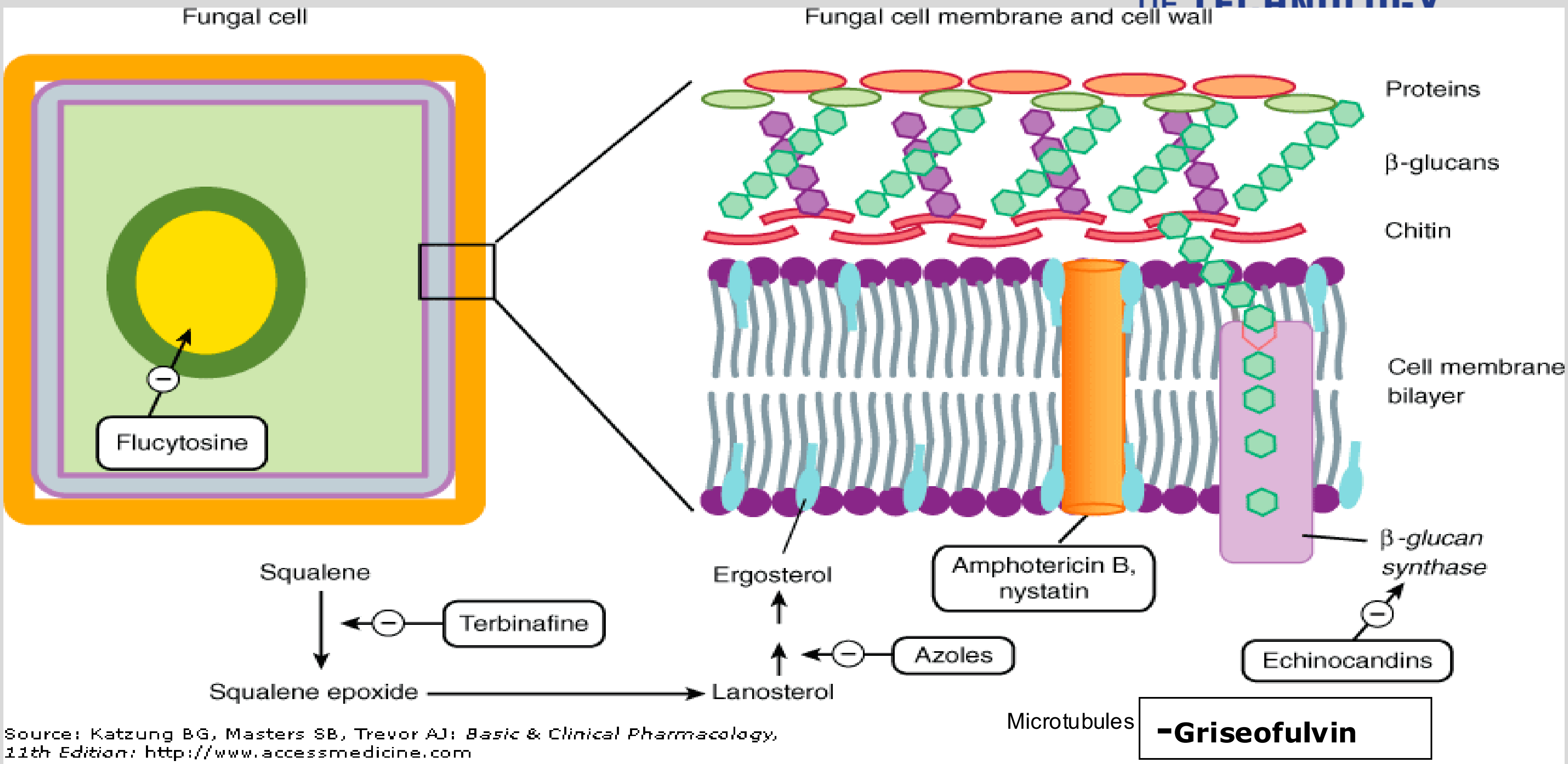
Source: Course Syllabus

Fungal Meningitis

- *Cryptococcus* (main cause)
- Treatment: flucytosine or fluconazole (voriconazole) with amphotericin.

Source: Course Syllabus

Antifungals



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

Amphotericin

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- Broad spectrum, fungicidal
- Intrathecal or intraventricular administration (fungal meningitis)
- <u>Resistance:</u> reduced cell membrane ergosterol
- <u>Adverse effects:</u> Infusion "shake and bake" reaction: prophylaxis (cyclooxygenase blockers)
- Renal tubule toxicity (Na, Ca, K, Mg excreted)
- Use liposomal preparations, give other antifungals to lessen toxicity. Hydration

Source: Course Syllabus

Azoles

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- Fluconazole (CSF)
- Voriconazole (systemic infections)
- 14α demethylase inhibitors, fungistatic
- Adverse effects: hepatic enzyme elevation
- <u>Inhibitors of CYP3A4</u>
- Ketoconazole (gynecomastia)

Source: Course Syllabus

Flucytosine

- Converted to 5-fluorouracil (5-FU)
- Inhibits DNA synthesis, pyrimidine analog, fungicidal
- Used with amphotericin to stop resistance against *Candida* and *Crypto*.
- Adverse effects: bone marrow suppression

Source: Course Syllabus

Summary of Current Empiric Therapy (IV)

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Neonate to 1 month old	Ampicillin and cefotaxime or gentamicin and acyclovir
Pediatric	Ampicillin and ceftriaxone
Adults < 50 years	Ceftriaxone and vancomycin
Adults > 50 years of age or immunosuppressed	Ceftriaxone and Vancomycin and Ampicillin
Meningitis due to trauma	Cefepime or ceftazidime or meropenem
Meningitis with penicillin allergy	Moxifloxacin and Vancomycin

https://www.ncbi.nlm.nih.gov/books/NBK459360/

Summary

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- **Describe the specific pharmacological treatments for bacterial, viral, and fungal meningitis.**
- **Describe the mechanisms of these drugs. Include adverse effects.**

Core References:

- **Katzung & Vanderah Basic and Clinical Pharmacology, 16e, 2024; Chapter 43, Beta-Lactam & Other Cell Wall- & Membrane-Active Antibiotics; Chapter 44, Chloramphenicol; Chapter 45, Aminoglycosides; Chapter 48, Antifungals; Chapter 49, Antivirals; Chapter 55, Glucocorticoids.**
- **CNS Infections Summary Guide and Practice Questions (Maria A. Pino, Ph.D)**

Source: Course Syllabus

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R